

**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****CERTIFICATE OF INSTALLATION****Note:** This table completed by ~~HERS~~ECC Registry.

Project Name:	Enforcement Agency:
Dwelling Address:	Permit Number:
City and Zip Code:	Permit Application Date:

Title 24, Part 6, Section 150.0(o) **Ventilation for Indoor Air Quality**. All dwelling units shall meet the requirements of ANSI/ASHRAE Standard 62.2-~~2019~~2022 Ventilation and Acceptable Indoor Air Quality in ~~Low-Rise Residential~~Single-Family Buildings subject to the amendments specified by Title 24, Part 6, Section 150.0(o)1

A. Whole-Dwelling Mechanical Ventilation - General Information**Note:**

Non-dwelling units do not meet the definition for a dwelling unit as defined in Section 100.1(b). Non-dwelling units are not designed to provide independent living facilities and do not provide permanent provisions for living, sleeping, eating, cooking and sanitation.

01	Dwelling Unit Name	
02	Building Type	
03	Project Scope	
04	Total Conditioned Floor Area of Dwelling Unit (For addition projects the conditioned floor area equals existing area plus addition area)	
05	Number of Bedrooms in Dwelling Unit (For addition projects the number of bedrooms equals the existing bedrooms plus addition bedrooms)	
06	Ventilation System Type	
07	Ventilation Operation Schedule	
08	<u>Fault Indicator Display (FID) Status</u>	

MCH-27a – Single Family Attached/Detached Ventilation**B1. Non-Dwelling Unit Compliance Statement**

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(Or Sections B through K for dwelling unit projects)

B. Single Family Attached/Detached General Information

01	Average Ceiling Height	
02	Total Conditioned Volume	
03	Vertical distance between the lowest and highest above-grade points within the pressure boundary in feet	
04	Air Changes Per Hour at 50 Pa	
05	Name of ANSI/ASHRAE Standard 62.2- 2019 <u>2022</u> weather station for climate zone	
06	Weather and shielding factor (wsf) (Based on the city identified above)	

**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****C. Ventilation - Total Ventilation Rate**

A mechanical supply system, exhaust system, or combination thereof shall provide whole-dwelling ventilation with outdoor air each hour at no less than the rate in 150.0(o)1Ci

01	Total Required Ventilation rate, (Q_{tot})	
02	Enclosure Leakage Rate (Q_{50})	
03	Effective Annual Average Infiltration Rate (Q_{inf})	
04	Total Exterior Envelope Surface Area	
05	Unshared Exterior Envelope Surface Area (exclude surface areas attached to garages or other dwelling units)	
06	Required Mechanical Ventilation Rate (Q_{fan})	

D. Installed Ventilation - Total Ventilation Rate

A mechanical supply system, exhaust system, or combination thereof shall provide whole-dwelling ventilation with outdoor air each hour at no less than the rate in 150.0(o)1Ci

01	02	03	04	05
Fan Name	Fan Location	Runtime (Min/Hr)	Installed Mechanical Ventilation Rate (CFM)	Equivalent Continuous Ventilation (CFM)
06	Total Installed Equivalent Continuous Ventilation (CFM)			

D2E. HRV or ERV Serving Individual Dwelling Unit Information

Balanced ventilation systems shall comply with appropriate requirements in 150.0(o)2C.

01	02	03
Manufacturer Make	Manufacturer Model Number	Fan Efficacy Performance Rating (W/CFM)

F. Requirements for balanced and supply only ventilation systems (150.0(o)Civa1)

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

Balanced and supply ventilation component accessibility. Balanced and supply ventilation systems shall meet the following requirements for accessibility:

01	IAQ filter and HRV/ERV accessibility. System air filters and HRV/ERV heat/energy recovery cores shall be located such that they are accessible for service from within occupiable spaces, basements, garages, balconies, and mechanical closets. Filters and heat/energy cores behind access panels, access doors, or grilles located no more than 10 feet above a walking surface inside a space specified above comply with this requirement. Exception to Section 150.0(o)1Civa: Systems that require servicing from inside the attic shall have the following: 1. A Fault Indicator Display (FID) meeting the requirements of Reference Appendix JA17; and 2. An attic access door located in a wall or, where attic access is provided through a ceiling, an attic access hatch that includes an integrated ladder; and 3. A walkway from the attic access door to the HRV/ERV.
02	IAQ System component accessibility: Fans, motors, heat exchangers, filters and recovery cores shall meet all applicable requirements of California Mechanical Code Section 304.0 accessibility for service.
03	Outdoor air intake design: Outdoor air intakes shall comply with California Mechanical Code Section 402.4.1.

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04	Outdoor air intake location and accessibility: To provide access for cleaning, outdoor air intakes shall be accessible. Air intakes located not more than 10 feet above a walking surface comply with this requirement. If located on roofs, they shall meet the requirements of California Mechanical Code Section 304.3.1. Exception to Section 150.0(o)1Civd: Outdoor air intake serving equipment with an FID meeting requirement of Reference Appendix JA 17.
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G. Fault Indicator Display

Qualification Requirements ~~Procedures~~ for Ventilation System ~~the~~ Fault Indicator Displays ~~Verification~~ are detailed in in Appendix JA17.

01	FID Manufacturer Name/Make	
02	FID Model Number	
03	The display module is mounted adjacent to the system thermostat.	
04	The manufacturer has certified to the Energy Commission that the FID model meets the requirements of Reference Joint Appendix JA17 (make and model found on CEC list of approved FID devices).	
05	The system has operated for at least 15 minutes and the FID reports that the system is operating within acceptable parameters.	

H. Fault Indicator Display – Additional Requirements

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

The responsible persons signature on this document indicates the installation complies with the following requirements:

01	Fault Indicator Display devices shall either be factory installed by the space-conditioning system manufacturer, or field installed according to the space-conditioning system manufacturer's requirements and the FID manufacturer's specifications.
02	The installer shall ensure that a copy of the FID manufacturer's user instructions documentation has been made available to the building owner.

I. Dwelling Unit Compliance Statement

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**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****FJ. Other Requirements**

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

The items listed below (6.1 through 6.6 and 6.8 ~~through 6.9~~) correspond to the information given in ASHRAE 62.2 Section 6 "Other Requirements". Refer also to Chapter 4.6 of the Residential Compliance Manual (Section 4.6.8) for information describing these "Other Requirements". The signature of the Responsible Person in the declaration statement below certifies that the dwelling complies with these requirements specified in ASHRAE 62.2 Section 6.1 through 6.6 and 6.9 if applicable.

01	6.1 Adjacent Spaces and Transfer Air. Measures shall be taken to minimize air movement across envelope components to dwelling units from adjacent spaces such as garages, unconditioned crawlspaces, unconditioned attics, and other dwelling. Supply and balanced <u>mechanical</u> ventilation systems shall be designed and constructed to provide ventilation air directly from the outdoors. 6.1.1 Compliance for Attached Dwelling Units. Attached dwelling units, except existing units as described in Normative Appendix A, Section A5, shall demonstrate compliance with Section 6.1 by verifying a leakage rate less than or equal to 0.2 cfm per ft² (100 L/s per 100 m²) of the dwelling-unit boundary area by means of a blower door test at a test pressure of 50 Pa. Testing shall be conducted in accordance with ANSI/RESNET/ICC Standard 380. For horizontally attached dwelling units that are being evaluated for the infiltration credit in Section 4.1.2, the procedure specified in Section 4.1.2 shall be an alternative to the procedure of this section. One method of demonstrating compliance with Section 6.1 shall be to verify a leakage rate below a maximum of 0.3 cfm per ft² (150 L/s per 100 m²) of the dwelling unit envelope area (i.e., the sum of the area of walls between dwelling units, exterior walls, ceiling, and floor) at a test pressure of 50 Pa by a blower door test conducted in accordance with either ANSI/ASTM E779 or ANSI/ASTM E1827. The test shall be conducted with the dwelling unit as if it were exposed to outdoor air on all sides, top, and bottom by opening doors and windows of adjacent dwelling units. 6.1.2 Garages. When an occupiable space adjoins a garage, the design must prevent migration of contaminants to the adjoining occupiable space. Air seal the walls, ceilings, and floors that separate garages from occupiable space. To be considered air-sealed, all joints, seams, penetrations, openings between door assemblies and their respective jambs and framing, and other sources of air leakage through wall and ceiling assemblies separating the garage from the residence and its attic area shall be caulked, gasketed, weather stripped, wrapped, or otherwise sealed to limit air movement. Doors between garages and occupiable spaces shall be gasketed or made substantially airtight with weather stripping. 6.1.3 Space-Conditioning System Ducts. All air distribution joints located outside the dwelling-unit boundary shall be sealed. HVAC systems that serve spaces within the dwelling-unit boundary shall not be designed to supply air to or return air from the garage. HVAC systems that include air handlers or ducts located outside the dwelling-unit boundary shall have total air leakage of no more than 6% of total fan airflow when measured at 0.1 in. of water (25 Pa) using California Building Energy Efficiency Standards, Residential Appendix RA3.1 or equivalent. Method D of ASTM E1554 may be used to meet this requirement. If the air handler, ducts, or both are located in the garage, the garage door shall be open to the outside when the duct leakage is tested.
02	6.2 Instructions and Labeling Controls shall be labeled as to their function (unless that function is obvious, such as toilet exhaust fan switches). Information on the ventilation design and/or ventilation systems installed, instructions on their proper operation to meet the requirements of this standard, and instructions detailing any required maintenance (similar to that provided for HVAC systems) shall be provided to the owner and the occupant of the dwelling unit. Controls shall be labeled as to their function (unless that function is obvious, such as toilet exhaust fan switches). See Section 13 of ASHRAE Guideline 24-5 for information on instructions and labeling.
03	6.3 Clothes Dryers. Clothes dryers shall be exhausted directly to the outdoors. Exception to 6.3: Condensing dryers plumbed to a drain.
04	6.4 Combustion and Solid-Fuel Burning Appliances. 6.4.1 Combustion and solid-fuel-burning appliances must be provided with adequate combustion and ventilation air and installed in accordance with manufacturers' installation instructions, NFPA 31, NFPA 54/ANSI Z223.1, NFPA 211, -or other equivalent code acceptable to the building official. 6.4.2 Where atmospherically vented combustion appliances or solid-fuel burning appliances are located inside the pressure dwelling unit boundary, the total net exhaust flow of the two largest exhaust fans (not including a summer cooling fan intended to be operated in conjunction with only when windows or other air inlets are open openings) shall not exceed 15 cfm per 100 ft² (75 L/s per 100 m²) of occupiable space floor area -when in operation at full capacity. If the designed total net airflow exceeds this limit, the net exhaust air flow must be reduced by reducing the exhaust air flow or providing compensating outdoor air. Gravity or barometric dampers in nonpowered exhaust makeup air systems shall not be used to provide compensating outdoor air. Atmospherically vented combustion appliances do not include direct-vent appliances.

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	Combustion appliances that pass safety testing performed according to ANSI/BPI-1200 shall be deemed as complying with Section 6.4.2.
05	6.5 Air-tightness Requirements 6.5.1 Garages. When an occupiable space adjoins a garage, the design must prevent migration of contaminants to the adjoining occupiable space. Air seal the walls, ceilings, and floors that separate garages from occupiable space. To be considered air-sealed, all joints, seams, penetrations, openings between door assemblies and their respective jambs and framing, and other sources of air leakage through wall and ceiling assemblies separating the garage from the residence and its attic area shall be caulked, gasketed, weather-stripped, wrapped, or otherwise sealed to limit air movement. Doors between garages and occupiable spaces shall be gasketed or made substantially airtight with weather stripping.
0605	6.6.5 Ventilation Opening Area. Spaces shall have ventilation openings as listed below in the following subsections. Such openings shall meet the requirements of Section 6.86. Exception to 6.5: Attached dwelling units and spaces that meet the local ventilation requirements set for bathrooms in Section 5 [of ASHRAE 62.2]. 6.6.5.1 Habitable Spaces. Each habitable space shall be provided with ventilation openings with an openable area not less than 4% of the floor area or less than 5 ft² (0.5 m²). 6.6.5.2 Toilets and Utility Rooms. Toilets and utility rooms shall be provided with <u>natural</u> ventilation openings with an openable area not less than 4% of the room floor area or less than 1.5 ft² (0.15 m²). Exceptions to 6.5.2: 1. Utility rooms with a dryer exhaust duct. 2. Toilet compartments in bathrooms.
0706	6.86.6 Air Inlets. Air inlets that are part of the ventilation design shall be located a minimum of 10 ft (3 m) from known sources of contamination such as a stack, vent, exhaust hood, or vehicle exhaust. The intake shall be placed so that entering air is not obstructed by snow, plantings, or other material. Forced air inlets shall be provided with rodent/insect screens (mesh not larger than 1/2 0.5 in. [13 mm]). Exceptions to 6.6: 1. Ventilation openings in the wall may be as close as a stretched-string distance of 3 ft (1 m) from sources of contamination exiting through the roof or dryer exhausts. 2. No minimum separation distance shall be required between windows and local exhaust outlets in kitchens and bathrooms. 3. Vent terminations covered by and meeting the requirements of the <i>National Fuel Gas Code</i> (NFPA 54/ANSI Z223.1) 7 or equivalent. 4. Where a combined exhaust/intake termination is used to separate intake air from exhaust air originating in a living space other than kitchens, no minimum separation distance between these two openings is required. For these combined terminations, the exhaust air concentration within the intake airflow shall not exceed 10%, as established by the manufacturer.
0807	6.96.8 Carbon Monoxide Alarms. A carbon monoxide alarm shall be installed in each dwelling unit in accordance with NFPA 720, Standard for the Installation of Carbon Monoxide (CO) Detection and Warning Equipment <u>National Fire Alarm and Signaling Code</u>, and shall be consistent with requirements of applicable laws, codes, and standards.
0908	Air Filter Efficiency. Supply only ventilation systems, makeup air-systems, and supply side balanced systems including HRV/ERV shall be provided with air filters having a designated efficiency equal to or greater than MERV 13 when tested in accordance with ASHRAE Standard 52.2, or a particle size efficiency rating equal to or greater than 50% in the 0.30-1.0 µm range and equal to or greater than 85% in the 1.0-3.0 µm range when tested in accordance with AHRI Standard 680.

**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****GK. Air Moving Equipment**

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

The items listed below (7.1 through **7.47.5**) correspond to the information given in ASHRAE 62.2 Section 7 "Air-Moving Equipment". Refer also to Chapter 4.6 of the Residential Compliance Manual (Section 4.6.9) for information describing these requirements in more detail. The signature of the Responsible Person in the declaration statement below certifies that the dwelling complies with these requirements specified in ASHRAE 62.2 Section 7.1 through **7.47.5** if applicable.

01	7.1 Selection and Installation Ratings. Ventilation devices and equipment serving individual dwelling units shall be tested in accordance with ANSI/ASHRAE Standard 51/AMCA 210, and ANSI/AMCA Standard 300, and rated in accordance with the airflow and sound rating procedures of the Home Ventilating Institute (HVI 915, 916, 920) Installations of systems or equipment shall be carried out in accordance with manufacturers' design requirements and installation instructions. Airflow and sound ratings shall be provided for ventilation devices and equipment serving individual dwelling units. Airflow and sound ratings shall be provided in accordance with HVI 920, or equivalent, by an administration and certification body that is accredited in accordance with ISO/IEC 17065 with respect to application of the standards and test procedures referenced in Section 7.1 and accredited by an accreditation body operating in accordance with ISO/IEC 17011. Laboratory tests of representative units shall be conducted for airflow in accordance with ANSI/ASHRAE Standard 51/AMCA 210, as prescribed by HVI 916, or equivalent, and conducted for sound in accordance with ANSI/AMCA Standard 300, as prescribed by HVI 915, or equivalent. This section does not require certification to HVI 917
02	7.2 Installation. Installations of systems or equipment shall be carried out in accordance with manufacturer's design requirements and installation instructions.
0203	7.27.3 Sound Ratings for Fans. Ventilation fans shall be rated for sound at no less than the minimum airflow rate required by this standard as noted below. These sound ratings shall be at a minimum of 0.1 in. of water (25 Pa) static pressure in accordance with the HVI procedures referenced in Section 7.1. Exception to 7.3: HVAC air handlers and remote mounted fans need not meet sound requirements. To be considered for this exception, a remote mounted fan must be mounted outside the habitable spaces, bathrooms, toilets, and hallways, and there must be at least 4 ft (1 m) of ductwork between the fan and the intake grille. 7.23.1 Dwelling-Unit Ventilation or Continuous Local Exhaust Fans. These fans shall be rated for sound at a maximum of 1.0 sone. 7.23.2 Demand-Controlled Local Exhaust Fans. Bathroom exhaust fans used to comply with Section 5.2 shall be rated for sound at a maximum of 3 sones. Kitchen exhaust fans used to comply with Section 5.2 shall be rated for sound at a maximum of 3 sones at one or more airflow settings greater than or equal to 100 cfm (47 L/s). Exceptions to 7.3.2: 1- Fans with a minimum airflow setting exceeding 400 cfm (189 L/s) need not comply. 2- Kitchen Range hoods may be rated for sound at the static pressure determined at working speed as specified in HVI 916 section 7.
0304	7.37.4 Exhaust Ducts. 7.34.1 Multiple Exhaust Fans Using One Duct. Exhaust fans in separate dwelling units shall not share a common exhaust duct. If more than one of the exhaust fans in a single dwelling unit shares a common exhaust duct, each fan shall be equipped with a backdraft damper to prevent the recirculation of exhaust air from one room to another through the exhaust ducting system. 7.34.2 Single Exhaust Fan Ducted to Multiple Exhaust Inlets. Where exhaust inlets are commonly ducted across multiple dwelling units, one or more exhaust fans located downstream of the exhaust inlets shall be designed and intended to run continuously, or a system of one or more backdraft dampers shall be installed to isolate each dwelling unit from the common duct when the fan is not running.
0405	7.47.5 Supply Ducts. Where supply outlets are commonly ducted across multiple dwelling units, one or more supply fans located upstream of all the supply outlets shall be designed and intended to run continuously, or a system of one or more backdraft dampers shall be installed to isolate each dwelling unit from the common duct when the fan is not running.

**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****DOCUMENTATION AUTHOR'S DECLARATION STATEMENT**

1. I certify that this Certificate of Installation documentation is accurate and complete.

Documentation Author Name:	Documentation Author Signature:
Documentation Author Company Name:	Date Signed:
Address:	CEA/ HERS ECC Certification Identification (If applicable):
City/State/Zip:	Phone:

RESPONSIBLE PERSON'S DECLARATION STATEMENT

I certify the following under penalty of perjury, under the laws of the State of California:

1. The information provided on this certificate of installation is true and correct.
2. I am either: a) a responsible person eligible under division 3 of the business and professions code in the applicable classification to accept responsibility for the system design, construction, or installation of features, materials, components, or manufactured devices for the scope of work identified on this certificate of installation, and attest to the declarations in this statement, or b) I am an authorized representative of the responsible person and attest to the declarations in this statement on the responsible person's behalf.
3. The constructed or installed features, materials, components or manufactured devices (the installation) identified on this certificate of installation conforms to all applicable codes and regulations and the installation conforms to the requirements given on the certificate of compliance, plans, and specifications approved by the enforcement agency.
4. I understand that a ~~HERS~~ECC rater will check the installation to verify compliance and if such checking determines the installation fails to comply, I am required to offer any necessary corrective action at no charge to the building owner.
5. I understand that a registered copy of this certificate of installation shall be posted or made available with the building permit(s) issued for the building and made available to the enforcement agency for all applicable inspections, and I will take the necessary steps to ensure this requirement is accomplished.
6. I understand that a registered copy of this certificate of installation is required to be included with the documentation the builder provides to the building owner at occupancy, and I will take the necessary steps to ensure this requirement is accomplished.

Responsible Builder/Installer Name:	Responsible Builder/Installer Signature:	
Company Name: (Installing Subcontractor or General Contractor or Builder/Owner)	Position With Company (Title):	
Address:	CSLB License:	
City/State/Zip:	Phone:	Date Signed:

CF2R-MCH-27-H User Instructions

Section A. General Information

1. Building Unit Name: This field is filled out automatically. It is referenced from the CF2R-MCH-01, which must be completed prior to this document. This is the unique identifier for this dwelling unit. Ventilation is calculated and provided for each dwelling unit individually.
2. Building Type: This field is filled out automatically. It is referenced from the CF1R. Values are “Single Family Attached” and “Single Family Detached”. User is allowed to overwrite imported value with “Non-dwelling unit” selection.
3. Project Scope: This field is filled out automatically. It is referenced from the CF1R.
 - If parent document is the CF1R-PRF-01, values are “Newly Constructed”, “Newly Constructed (Addition Alone)” and “Addition and /or Alteration”
 - If parent document is CF1R-NCB-01, values are “Newly Constructed” and “Newly Constructed (Addition Alone)”
 - If parent document is CF1R-ADD-01, values are “ADU Addition < 300 ft²”, “ADU Addition > 300 to < 400 ft²”, “ADU Addition > 400 to < 700 ft²” and “ADU Addition > 700 to < 1000 ft²”.
4. Total Conditioned Floor Area of Dwelling Unit: This field is filled out automatically. It is referenced from the CF2R-MCH-01.
5. Number of Bedrooms in Dwelling Unit: This field is filled out automatically. It is referenced from the CF2R-MCH-01.
6. Ventilation system Type: This may be filled out automatically or be user input.
 - If parent document is the CF1R-PRF-01, the value will be filled out automatically.
 - If building type is equal to Non-dwelling unit, an N/A value will be filled out automatically.
 - If parent document is the CF1R-NCB or CF1R-ADD, user selects from list of Supply, Exhaust, Balanced, Balanced – ERV, Balanced – HRV, Central Fan Integrated (CFI), Central Ventilation System – Supply and Central Ventilation System – Exhaust and Central Ventilation System Balanced.
7. Ventilation operation schedule: This may be filled out automatically or be user input.
 - Building type is equal to Non-dwelling unit; an N/A value will be filled out automatically.
 - User selects from list of Continuous, Short-Term Average, Scheduled and Real-time Control.
 - Note if “Ventilation System Type” (A06) = Central Fan Integrated & “Ventilation Operation Schedule” (A07) = Continuous; then user will not be allowed to proceed.

Section B1. Non-Dwelling Unit Compliance Statement

If this is a non-dwelling unit Section B1 is the only section besides Section A in the report.

1. This is automatically completed.

If this is a single family detached or single family attached, the following Sections B through K are required

Section B. Single Family Attached/Detached General Information

1. Average Ceiling Height: This may be filled out automatically or be user input.
 - If parent document is the CF1R-PRF-01, the value will be filled out automatically.
 - If parent document is the CF1R-NCB or CF1R-ADD, user enter value in feet.
2. Total Conditioned Volume: This field is calculated and filled out automatically.
3. Vertical distance between the lowest and highest above-grade points within the pressure boundary in feet: This may be filled out automatically or be user input.
 - If parent document is the CF1R-PRF-01, the value will be filled out automatically.

- If parent document is the CF1R-NCB or CF1R-ADD, user enters value in feet.
- 4. Air Changes Per Hour at 50 Pa: This may be filled out automatically or be user selected
 - If Building type is equal to Non-dwelling unit, an N/A value will be filled out automatically.
 - If Building type does not equal Non-dwelling unit, then user may select from Default (ACH50=2.0) or Measured (ACH50<2.0)
- 5. Name of ANSI/ASHRAE Standard 62.2-2016 weather station for climate zone: This may be filled out automatically or be user input.
 - If parent document is the CF1R-PRF-01, the value will be filled out automatically.
 - If Building type is equal to Non-dwelling unit, an N/A value will be filled out automatically.
 - If parent document is the CF1R-NCB or CF1R-ADD, user select value from Weather Stations from the Table X1 US Climates, Normative Appendix X.
- 6. Weather and shielding factor (wsf): This value is automatically entered based on the selection in #6.

Section C. Whole-Dwelling Continuous Ventilation – Total Ventilation Rate Method

1. This value is automatically calculated using equation 150.0-B from the Energy Standards.
2. This value automatically calculates using either equation 150.0-C or 150.0-D from the Energy Standards.
 - If air changes per hour from section B is equal to “Default” then equation, 150.0-C will be used.
 - If air changes per hour from section B is equal to “Measured” and the leakage value from the CF2R-MCH-24 is < 2.0 then equation 150.0-D will be used.
 - If air changes per hour from section B is equal to “Measured” and the leakage value from the CF2R-MCH-24 is ≥ 2.0 then equation 150.0-C will be used.
3. This value is automatically calculated using equation 150.0-E from the Energy Standards.
4. Total Exterior Envelope Surface Area: This value may be filled out automatically or be user input.
 - If dwelling type from section A equals “Single Family Detached”, an N/A value will be filled out automatically.
 - If dwelling type from section A equals “Single Family Attached” and the parent document is the CF1R-PRF-01 then value will be automatically entered.
 - If dwelling type from section A equals “Single Family Attached” and the parent document is the CF1R-NCB-01 or CF1R-ADD-01 then user enter value (ft²).
5. Unshared Exterior Surface Area: This value may be filled out automatically or be user input.
 - If dwelling type from section A equals “single family detached”, an N/A value will be filled out automatically.
 - If dwelling type from section A equals “single family attached” and the parent document is the CF1R-PRF-01 then value will be automatically entered.
 - If dwelling type from section A equals “single family attached” and the parent document is the CF1R-NCB-01 or CF1R-ADD-01 then user enter value (ft²).
6. This value is automatically calculated using equation 150.0-F from the Energy Standards.

Section D. Installed Ventilation – Total Ventilation Rate Method

1. User input text identifying the fan name for each installed ventilation fan.
2. User input text identifying the fan location for each installed ventilation fan.
3. Runtime (Min/Hr): This value may be filled out automatically or be user input.
 - If ventilation operation schedule from section B = “continuous”, then value of 60 will be automatically entered.

- If ventilation operation schedule from section B = “short term average”, then user enter value of less than or equal to 60 for each installed ventilation fan.
4. User to enter CFM value from test procedures described in RA3.7.4 for each installed ventilation fan.
 5. Equivalent continuous ventilation CFM is automatically calculated for each ventilation fan.
 6. Total installed equivalent continuous ventilation CFM is automatically calculated based on the installed ventilation fans.

NORMATIVE APPENDIX B:
INFILTRATION EFFECTIVENESS WEATHER AND SHIELDING FACTORS (WSF)
TABLE B1 U.S. Climates

TMY3	wsf	Weather Station	Latitude	Longitude	State
690150	0.50	Twentynine Palms	34.30	–116.17	California
722860	0.43	March AFB	33.90	–117.25	California
722868	0.45	Palm Springs Intl	33.83	–116.50	California
722869	0.42	Riverside Muni	33.95	–117.45	California
722880	0.39	Burbank–Glendale–Pasadena AP	34.20	–118.35	California
722885	0.39	Santa Monica Muni	34.02	–118.45	California
722886	0.39	Van Nuys Airport	34.22	–118.48	California
722895	0.55	Lompoc (AWOS)	34.67	–120.47	California
722897	0.51	San Luis Co Rgnl	35.23	–120.63	California
722899	0.45	Chino Airport	33.97	–117.63	California
722900	0.38	San Diego Lindbergh Field	32.73	–117.17	California
722903	0.39	San Diego/Montgomery	32.82	–117.13	California
722904	0.40	Chula Vista Brown Field NAAS	32.58	–116.98	California
722906	0.39	San Diego North Island NAS	32.70	–117.20	California
722926	0.40	Camp Pendleton MCAS	33.30	–117.35	California
722927	0.38	Carlsbad/Palomar	33.13	–117.28	California
722930	0.39	San Diego Miramar NAS	32.87	–117.13	California
722950	0.42	Los Angeles Intl Arpt	33.93	–118.40	California
722956	0.38	Jack Northrop Fld H	33.92	–118.33	California
722970	0.38	Long Beach Daugherty Fld	33.83	–118.17	California
722976	0.34	Fullerton Municipal	33.87	–117.98	California
722977	0.36	Santa Ana John Wayne AP	33.68	–117.87	California
723805	0.51	Needles Airport	34.77	–114.62	California
723810	0.59	Edwards AFB	34.90	–117.87	California
723815	0.58	Daggett Barstow–Daggett AP	34.85	–116.80	California
723816	0.62	Lancaster Gen Wm Fox Field	34.73	–118.22	California
723820	0.57	Palmdale Airport	34.63	–118.08	California
723830	0.68	Sandberg	34.75	–118.72	California
723840	0.43	Bakersfield Meadows Field	35.43	–119.05	California
723890	0.45	Fresno Yosemite Intl AP	36.78	–119.72	California
723895	0.42	Porterville (AWOS)	36.03	–119.07	California
723896	0.43	Visalia Muni (AWOS)	36.32	–119.40	California
723910	0.45	Point Mugu Nf	34.12	–119.12	California

NORMATIVE APPENDIX B:
INFILTRATION EFFECTIVENESS WEATHER AND SHIELDING FACTORS (WSF)

TABLE X1 U.S. Climates

TMY3	wsf	Weather Station	Latitude	Longitude	State
723925	0.44	Santa Barbara Municipal AP	34.43	-119.85	California
723926	0.43	Camarillo (AWOS)	34.22	-119.08	California
723927	0.45	Oxnard Airport	34.20	-119.20	California
723940	0.52	Santa Maria Public Arpt	34.92	-120.47	California
723965	0.53	Paso Robles Municipal Arpt	35.67	-120.63	California
724800	0.55	Bishop Airport	37.37	-118.35	California
724815	0.46	Merced/Macready Fld	37.28	-120.52	California
724830	0.51	Sacramento Executive Arpt	38.50	-121.50	California
724837	0.45	Beale AFB	39.13	-121.43	California
724838	0.50	Yuba Co	39.10	-121.57	California
724839	0.51	Sacramento Metropolitan AP	38.70	-121.58	California
724915	0.49	Monterey Naf	36.60	-121.87	California
724917	0.54	Salinas Municipal AP	36.67	-121.60	California
724920	0.50	Stockton Metropolitan Arpt	37.90	-112.23	California
724926	0.47	Modesto City – County AP	37.63	-120.95	California
724927	0.53	Livermore Municipal	37.70	-121.82	California
724930	0.54	Oakland Metropolitan Arpt	37.72	-122.22	California
724935	0.47	Hayward Air Term	37.67	-122.12	California
724936	0.53	Concord – Buchanan Field	38.00	-122.05	California
724940	0.60	San Francisco Intl AP	37.62	-122.40	California
724945	0.48	San Jose Intl AP	37.37	-121.93	California
724955	0.55	Napa Co. Airport	38.22	-122.28	California
724957	0.49	Santa Rosa (AWOS)	38.52	-122.82	California
725845	0.44	Blue Canyon AP	39.30	-120.72	California
725846	0.66	Truckee–Tahoe	39.32	-120.13	California
725847	0.64	South Lake Tahoe	38.90	-120.00	California
725905	0.47	Ukiah Municipal AP	39.13	-123.20	California
725910	0.50	Red Bluff Municipal Arpt	40.15	-122.25	California
725920	0.47	Redding Municipal Arpt	40.52	-122.32	California
725945	0.56	Arcata Airport	40.98	-124.10	California
725946	0.60	Crescent City Faa Ai	41.78	-124.23	California
725955	0.55	Montague Siskiyou County AP	41.78	-122.47	California
725958	0.59	Alturas	41.50	-120.53	California
745090	0.45	Mountain View Moffett Fld NAS	37.40	-122.05	California
745160	0.67	Travis Field AFB	38.27	-121.93	California
746120	0.52	China Lake Naf	35.68	-117.68	California
747020	0.50	Lemoore Reeves NAS	36.33	-119.95	California
747185	0.46	Imperial	32.83	-115.58	California
747187	0.46	Palm Springs Thermal AP	33.63	-116.17	California
747188	0.48	Blythe Riverside Co Arpt	33.62	-114.72	California

Section D2E. HRV or ERV Information

1. Manufacturer Make – User input text
2. Manufacturer Model Number – User input text

3. Fan Efficacy Performance Rating – Reference information from CF1R or be user input

Section F. Requirements for balanced and supply only ventilation systems

Section G. Fault Indicator Display

1. Enter the manufacturer name or make of the approved Fault Indicator Display. Must match name shown on the list of approved devices kept by the Commission.
2. Enter the manufacturer model number of the approved Fault Indicator Display. Must match name shown on the list of approved devices kept by the Commission.
3. The installer must confirm that the FID display module is mounted adjacent to thermostat that controls the system being verified. This requirement is detailed in Residential Appendix JA17.
4. The installer must confirm that the installed FID is approved and appears the list of approved devices kept by the Commission. This requirement is detailed in Residential Appendix JA17.
5. The installer must confirm that the system has operated for at least 15 minutes and that they system is operating within acceptable parameters as specified by the FID and equipment manufacturers. This requirement is detailed in Residential Appendix JA17.

Section H. Fault Indicator Display – Additional Requirements

Section I. Compliance Statement

1. Compliance Statement: This field is filled out automatically

Section J - Other Requirements

Section K - Air Moving Equipment

Title 24, Part 6, Section 150.0(o) **Ventilation for Indoor Air Quality**. All dwelling units shall meet the requirements of ANSI/ASHRAE Standard 62.2-2016 Ventilation and Acceptable Indoor Air Quality in **Low-Rise Single Family** Residential Buildings subject to the amendments specified by Title 24, Part 6, Section 150.0(o)1

A. Whole-Dwelling Mechanical Ventilation - General Information

01	Dwelling Unit Name	<<calculated field, referenced data from MCH-01, "Dwelling Unit Name" (A01); else if information not available then allow user input>>
02	Building Type	<< calculated field, referenced data from CF1R, allowed values = single family detached, single family attached, or allow user to pick: **Non-dwelling unit>>
03	Project Scope	<< calculated field, referenced data from CF1R; If parent document is CF1R-PRF-01, allowed values = Newly Constructed, Newly Constructed (Addition Alone), and Addition and /or Alteration; Else if parent document is CF1R-NCB-01, allowed values = Newly Constructed and Newly Constructed (Addition Alone); Else if parent document is CF1R-ADD-01, allowed values = ADU Addition < 300 ft2, ADU Addition > 300 to < 400 ft2, ADU Addition > 400 to < 700 ft2, and ADU Addition > 700 to < 1000 ft2>>
04	Total Conditioned Floor Area of Dwelling Unit (For addition projects the conditioned floor area equals existing area plus addition area)	<<calculated field: referenced from MCH-01, "Dwelling Unit Total Conditioned Floor Area (ft ²)" (A03); else if information not available then allow user input >>
05	Number of Bedrooms in Dwelling Unit (For addition projects the number of bedrooms equals the existing bedrooms plus addition bedrooms)	<<calculated field: referenced from MCH-01, "Dwelling Unit Number of Bedrooms" (A09); if value from MCH-01 = 0 replace with 1; else if information not available then allow user input >>
06	Ventilation System Type	<< calculated value if registered CF1R form equals CF1R-PRF-01, reference data from CF1R; Else if registered CF1R form equals CF1R-NCB-01 or CF1R-ADD-01, user pick one from list: **Supply **Exhaust; or **Balanced; or **Balanced – ERV; or **Balanced – HRV; or **Central Fan Integrated (CFI); or **Central Ventilation System – Supply; or **Central Ventilation System – Exhaust; or **Central Ventilation System – Balanced; or Else if "Building Type" (A02) = "Non-dwelling unit" then value = N/A>>
07	Ventilation Operation Schedule	<< calculated value if registered CF1R form equals CF1R-PRF-01, reference data from CF1R; Else if "Building Type" (A02) = "Non-dwelling unit", then value = N/A; Else if registered CF1R form equals CF1R-NCB-01 or CF1R-ADD-01, user pick one from list: **Continuous; or **Short-Term Average; Else if "Ventilation System Type" (A06) = Central Fan Integrated & "Ventilation Operation Schedule" (A07)= Continuous; then display: "Central Fan Integrated Ventilation System Not Allowed to Operate Continuously - Do Not Proceed ">>
08	<u>Fault Indicator Display (FID) Status</u>	<<user pick one from list: This system has a factory installed FID; or This system has a field installed FID; or This system does not have a FID device installed>>
09	Climate Zone (this row is not visible to the user)	<<value from CF1R>>

Note:

Non-dwelling units do not meet the definition for a dwelling unit as defined in Section 100.1(b). Non-dwelling units are not designed to provide independent living facilities and do not provide permanent provisions for living, sleeping, eating, cooking and sanitation.

<<If A02 = Non-dwelling unit Section B1 is required and the only section displayed in report>>

B1. Non-Dwelling Unit Compliance Statement

01	<< If 'Building Type' (A02) = 'Non-dwelling Unit' then display text: "Building Passes">>
----	--

<<If A02 = single family detached, single family attached Sections B through K apply>>

B. Single Family Attached/Detached General Information

01	Average Ceiling Height	<< calculated field; If parent document is CF1R-PRF-01, reference value from CF1R; Else if parent document is CF1R-NCB-01 or CF1R-ADD-01, user to enter nonnegative number with up to two decimals places (X.XX); Else if "Building Type" (A02) = "Non-dwelling unit" then value = N/A>>
02	Total Conditioned Volume	<<calculated value, "Conditioned Floor Area (A04)" * "Averaged Ceiling Height" (B01); Else if "Building Type" (A02) = "Non-dwelling unit" then value = N/A>>
03	Vertical distance between the lowest and highest above-grade points within the pressure boundary in feet	<< calculated field; If parent document is CF1R-PRF-01, reference value from CF1R; Else if parent document is CF1R-NCB-01 or CF1R-ADD-01, user to enter nonnegative number with up to two decimals places (X.XX); Else if "Building Type" (A02) = "Non-dwelling unit" then value = N/A>>
04	Air Changes Per Hour at 50 Pa	<<user pick from list: **Default; **Measured; Else if "Building Type" (A02) = "Non-dwelling unit" then value = N/A>>
05	Name of ANSI/ASHRAE Standard 62.2-2019 weather station for climate zone	<<Calculated field: If parent document is CF1R-PRF-01, reference value from it but allow user to override it. If not overridden, result is stored in B05_WeatherStationANSI_ASHRAE; Else if user overrides CF1R-PRF-01 value Or parent is CF1R-NCB-01 or CF1F-ADD-01, user selects from list of weather stations based on value in A09_ClimateZone and this is stored in B05_WeatherStationCZ# where # is the climate zone; Else If A02_ResidentialBuildingType == NonDwellingUnit result = N/A>> stored in NotApplicableMessage>>
06	Weather and shielding factor (wsf) (Based on the city identified above)	<<Calculated field: If parent document is CF1R-PRF-01 And B05_WeatherStationANSI_ASHRAE has a value, reference value from CF1R-PRF-01; Else If B05_WeatherStationCZ# (# is the climate zone) has a value, based on weather station value in B05_WeatherStationCZ#, assign lookup wsf from Table B1 US Climates, Normative Appendix B shown in the instructions section of the layout document; Else If A02_ResidentialBuildingType == NonDwellingUnit result = N/A stored in NotApplicableMessage>>

C. Ventilation - Total Ventilation Rate

<<Static Text - see the top>>

01	Total Required Ventilation rate, (Q_{tot})	<<calculated field, numeric: (use equation 4.1a): $[(0.03 * \text{Total Conditioned Floor Area of Dwelling Unit (A04)}) + (7.5 * (A05 + 1))]$, (cfm).>>
02	Enclosure Leakage Rate (Q_{50})	<<calculated field, if "Air Changes Per Hour at 50 Pa" (B04) = default, then value = "Total Conditioned Volume" (B02)*2/60 (CFM); Else if "Air Changes Per Hour at 50 Pa" (B04) = measured, and if ACH50 from MCH-24 D03 > 2.0, then value = B02*2/60 (CFM); Else if "Air Changes Per Hour at 50 Pa" (B04) = measured, and if ACH50 from MCH-24 D03 ≤ 2.0 value = B02*ACH50 from MCH-24 D03/60>>
03	Effective Annual Average Infiltration Rate (Q_{inf})	<<calculated field, $0.052 * \text{"Enclosure Leakage Rate (Q}_{50}\text{)" (C02) * "Weather and shielding factor (wsf)" (B06) * ("Vertical distance from the lowest above-grade floor to the highest ceiling" (B03)/8.2)}^{0.4}$ >>
04	Total Exterior Envelope Surface Area	<< calculated field, if Building Type (A02) = Single family Detached then value equals N/A; Else if Building Type (A02) = Single family Attached and if parent document is CF1R-PRF-01, then use imported value from CF1R and allow user to overwrite; If the value is not available from the CF1R-PRF allow the user to input a value; Else if Building Type (A02) = Single family Attached and if parent document is CF1R-NCB-01 or CF1R-ADD-01, then user then user enter whole number value (ft^2)>>
05	Unshared Exterior Envelope Surface Area (exclude surface areas attached to garages or other dwelling units)	<< calculated field, if Building Type (A02) = Single family Detached then value equals N/A; Else if Building Type (A02) = Single family Attached and if parent document is CF1R-PRF-01, then use imported value from CF1R and allow user to overwrite; If the value is not available from the CF1R-PRF allow the user to input a value;Else if Building Type (A02) = Single family Attached and if parent document is CF1R-NCB-01 or CF1R-ADD-01, then user then user enter whole number value (ft^2)>>
06	Required Mechanical Ventilation Rate (Q_{fan})	<<calculated value, if "Building Type" (A02) = Single Family Detached and "Ventilation System Type" (A06) = Balanced, Balanced – ERV or Balanced - HRV, then value = "Total Required Ventilation rate" (C01) – $[1 * (\text{"Effective Annual Average Infiltration Rate" (C03) * 1})]$; Else if "Building Type" (A02) = Single Family Attached and "Ventilation System Type" (A06) = Balanced, Balanced – ERV or Balanced - HRV, then value = "Total Required Ventilation rate" (C01) – $\{1 * [(\text{"Effective Annual Average Infiltration Rate" (C03) * ("Unshared Exterior Envelope Surface Area" (C05) / "Total Exterior Envelope Surface Area" (C04))}]]\}$; Else if "Building Type" (A02) = Single Family Detached and "Ventilation System Type" (A06) = Supply, Exhaust, or Central Fan Integrated, then value = "Total Required Ventilation rate" (C01) – $[(\text{Effective Annual Average Infiltration Rate (C03) / "Total Required Ventilation rate" (C01)}) * (\text{"Effective Annual Average Infiltration Rate" (C03) * 1})]$; Else if "Building Type" (A02) = Single Family Attached and "Ventilation System Type" (A06) = Supply, Exhaust, or Central Fan Integrated, then value = "Total Required Ventilation rate" (C01) – $\{(\text{Effective Annual Average Infiltration Rate (C03) / "Total Required Ventilation rate" (C01)}) * [(\text{"Effective Annual Average Infiltration Rate" (C03) * ("Unshared Exterior Envelope Surface Area" (C05) / "Total Exterior Envelope Surface Area" (C04))}]]\}$ >>

D. Installed Ventilation - Total Ventilation Rate

<<Static Text - see the top>>

01	02	03	04	05
Fan Name	Fan Location	Runtime (Min/Hr)	Installed Mechanical Ventilation Rate (CFM)	Equivalent Continuous Ventilation (CFM)
<< user input, text>>	<<user input, text>>	<<calculated field: if value in "Ventilation Operation Schedule" (A07) equals Continuous, then value equals 60; Else if value in "Ventilation Operation Schedule" (A07) equals Short Term Average ,then user input value positive integer ≤ 60>>	<< user input, positive integer>>	<<calculated field, value = ("Runtime (Min/Hr)" (D03) * "Installed Mechanical Ventilation Rate (CFM)" (D04)) / 60 (CFM)>>
06	Total Installed Equivalent Continuous Ventilation (CFM)			<<calculated field, value = sum of values in column "Equivalent Continuous Ventilation (CFM)" (D05)>>

D2E. HRV or ERV Serving Individual Dwelling Unit Information

<<Static Text - see the top>>

<<If A06=Balanced HRV or Balanced ERV then display table; Else display the "section does not apply" message>>

01	02	03
Manufacturer Make	Manufacturer Model Number	Fan Efficacy Performance Rating (W/CFM)
<<user input, text>>	<<user input, text>>	<<Call information from CF1R-PRF: Else user input, text, Range check: User entered value must be ≤ 1.0>>

F. Requirements for balanced and supply only ventilation systems (150.0(o)Civa1)

<<If A06=Balanced HRV or Balanced ERV then display table; Else display the "section does not apply" message>>

<<Static Text - see the top>>

01	<u>IAQ filter and HRV/ERV accessibility. System air filters and HRV/ERV heat/energy recovery cores shall be located such that they are accessible for service from within occupiable spaces, basements, garages, balconies, and mechanical closets. Filters and heat/energy cores behind access panels, access doors, or grilles located no more than 10 feet above a walking surface inside a space specified above comply with this requirement.</u> <u>Exception to Section 150.0(o)1Civa:</u> Systems that require servicing from inside the attic shall have the following: — An Fault Indicator Display (FID) meeting the requirements of Reference Appendix JA17; and — An attic access door located in a wall or, where attic access is provided through a ceiling, an attic access hatch that includes an integrated ladder; and — A walkway from the attic access door to the HRV/ERV.
02	<u>IAQ System component accessibility: Fans, motors, heat exchangers, filters and recovery cores shall meet all applicable requirements of California Mechanical Code Section 304.0 accessibility for service.</u>
03	<u>Outdoor air intake design: Outdoor air intakes shall comply with California Mechanical Code Section 402.4.1.</u>
04	<u>Outdoor air intake location and accessibility: To provide access for cleaning, outdoor air intakes shall be accessible. Air intakes located not more than 10 feet above a walking surface comply with this requirement. If located on roofs, they shall meet the requirements of California Mechanical Code Section 304.3.1.</u> <u>Exception to Section 150.0(o)1Civd:</u> Outdoor air intake serving equipment with an FID meeting requirement of Reference Appendix JA 17.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

G. Fault Indicator Display

<<Static Text - see the top>>

<<If A08 = "This system has a factory installed FID"; or "This system has a field installed FID", then display this section>>

01	FID Manufacturer Name/Make	<<user entry, text field (must be on list of approved devices)>>
----	----------------------------	--

02	<u>FID Model Number</u>	<<user entry, text field (must be on list of approved devices)>>
03	<u>The display module is mounted adjacent to the system thermostat.</u>	<<User select from list: Yes or No; pass if confirmed, else do not proceed .if No then do not proceed>>
04	<u>The manufacturer has certified to the Energy Commission that the FID model meets the requirements of Reference Joint Appendix JA17 (make and model found on CEC list of approved FID devices).</u>	<<User selects from list: Yes or No; pass if confirmed, else do not proceed .if No then do not proceed>>
05	<u>The system has operated for at least 15 minutes and the FID reports that the system is operating within acceptable parameters.</u>	<<User select from list: Yes or No; pass if confirmed, else do not proceed .if No then do not proceed>>

H. Fault Indicator Display – Additional Requirements

<<If A08 = “This system has a factory installed FID”; or “This system has a field installed FID”, then display this section>>

<<Static Text - see the top>>

01	Fault Indicator Display devices shall either be factory installed by the space-conditioning system manufacturer, or field installed according to the space-conditioning system manufacturer's requirements and the FID manufacturer's specifications.
02	The installer shall ensure that a copy of the FID manufacturer's user instructions documentation has been made available to the building owner.
<u>The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.</u>	

EI. Compliance Statement

01	<< If 'Building Type' (A02) = 'Non-dwelling Unit' then display text: "Building Passes"; If the 'Total Installed Mechanical Ventilation (D06) ≥ Required Mechanical Ventilation Rate (C06), then display text: "Building Passes Mechanical Ventilation Rate Test" else display text: "Building Fails Mechanical Ventilation Rate Test">>
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FJ. Other Requirements

The items listed below (6.1 through 6.6 and 6.8 ~~through 6.9~~) correspond to the information given in ASHRAE 62.2 Section 6 "Other Requirements". Refer also to Chapter 4.6 of the Residential Compliance Manual (Section 4.6.8) for information describing these "Other Requirements". The signature of the Responsible Person in the declaration statement below certifies that the dwelling complies with these requirements specified in ASHRAE 62.2 Section 6.1 through 6.6 and 6.8 ~~6.9~~ if applicable.

<<Static Text - see the top>>

01	6.1 Adjacent Spaces and Transfer Air. Measures shall be taken to minimize air movement across envelope components to dwelling units from adjacent spaces such as garages, unconditioned crawlspaces, unconditioned attics, and other dwelling. Supply and balanced mechanical ventilation systems shall be designed and constructed to provide ventilation air directly from the outdoors. 6.1.1 Compliance for Attached Dwelling Units. Attached dwelling units, except existing units as described in Normative Appendix A, Section A5, shall demonstrate compliance with Section 6.1 by verifying a leakage rate less than or equal to 0.2 cfm per ft² (100 L/s per 100 m²) of the dwelling unit boundary area by means of a blower door test at a test pressure of 50 Pa. Testing shall be conducted in accordance with ANSI/RESNET/ICC Standard 380. For horizontally attached dwelling units that are being evaluated for the infiltration credit in Section 4.1.2, the procedure specified in Section 4.1.2 shall be an alternative to the procedure of this section. One method of demonstrating compliance with Section 6.1 shall be to verify a leakage rate below a maximum of 0.3 cfm per ft² (150 L/s per 100 m²) of the dwelling unit envelope area (i.e., the sum of the area of walls between dwelling units, exterior walls, ceiling, and floor) at a test pressure of 50 Pa by a blower door test conducted in accordance with either ANSI/ASTME779 or ANSI/ASTM E1827. The test shall be conducted with the dwelling unit as if it were exposed to outdoor air on all sides, top, and bottom by opening doors and windows of adjacent dwelling units. 6.1.2 Garages. When an occupiable space adjoins a garage, the design must prevent migration of contaminants to the adjoining occupiable space. Air seal the walls, ceilings, and floors that separate garages from occupiable space. To be considered air-sealed, all joints, seams, penetrations, openings between door assemblies and their respective jambs and framing, and other sources of air leakage through wall and ceiling assemblies separating the garage from the residence and its attic area shall be caulked, gasketed, weather stripped, wrapped, or otherwise sealed to limit air movement. Doors between garages and occupiable spaces shall be gasketed or made substantially airtight with weather stripping. 6.1.3 Space Conditioning System Ducts. All air distribution joints located outside the dwelling unit boundary shall be sealed. HVAC systems that serve spaces within the dwelling unit boundary shall not be designed to supply air to or return air from the garage. HVAC systems that include air handlers or ducts located outside the dwelling unit boundary shall have total air leakage of no more than 6% of total fan airflow when measured at 0.1 in. of water (25 Pa) using California Building Energy Efficiency Standards, Residential Appendix RA3.1 or equivalent. Method D of ASTM E1554 may be used to meet this requirement. If the air handler, ducts, or both are located in the garage, the garage door shall be open to the outside when the duct leakage is tested.
02	6.2 Instructions and Labeling. Controls shall be labeled as to their function (unless that function is obvious, such as toilet exhaust fan switches). Information on the ventilation design and/or ventilation systems installed, instructions on their proper operation to meet the requirements of this standard, and instructions detailing any required maintenance (similar to that provided for HVAC systems) shall be provided to the owner and the occupant of the dwelling unit. Controls shall be labeled as to their function (unless that function is obvious, such as toilet exhaust fan switches). See Section 13 of ASHRAE Guideline 24⁵ for information on instructions and labeling.
03	6.3 Clothes Dryers. Clothes dryers shall be exhausted directly to the outdoors. Exception to 6.3: Condensing dryers plumbed to a drain.
04	6.4 Combustion and Solid Fuel Burning Appliances. 6.4.1 Combustion and solid fuel burning appliances must be provided with adequate combustion and ventilation air and installed in accordance with manufacturers' installation instructions, NFPA 31, NFPA 54/ANSI Z223.1, or NFPA 211, or other equivalent code acceptable to the building official. 6.4.2 Where atmospherically vented combustion appliances or solid fuel burning appliances are located inside the dwelling unit pressure boundary, the total net exhaust flow of the two largest exhaust fans (not including a summer cooling fan intended to be operated in conjunction with only when windows or other air inlets are open) shall not exceed 15 cfm per 100 ft² (75 L/s per 100 m²) of occupiable space floor area when in operation at full capacity. If the designed total net airflow exceeds this limit, the net exhaust airflow must be reduced by reducing the exhaust airflow or providing compensating outdoor air. Gravity or barometric dampers in nonpowered exhaust makeup air systems shall not be used to provide compensating outdoor air. Atmospherically vented combustion appliances do not include direct vent appliances. Combustion appliances that pass safety testing performed according to ANSI/BPI-1200 shall be deemed as complying with Section 6.4.2.
05	6.5 Air tightness Requirements 6.5.1 Garages. When an occupiable space adjoins a garage, the design must prevent migration of contaminants to the adjoining occupiable space. Air seal the walls, ceilings, and floors that separate garages from occupiable space. To be considered air-sealed, all joints, seams, penetrations, openings between door assemblies and their respective jambs and framing, and other sources of air leakage through wall and ceiling assemblies separating the garage from the residence and its attic area shall be caulked, gasketed, weather stripped, wrapped, or otherwise sealed to limit air movement. Doors between garages and occupiable spaces shall be gasketed or made substantially airtight with weather stripping.

0605	6.65.5 Ventilation Opening Area. Spaces shall have ventilation openings as listed below in the following subsections. Such openings shall meet the requirements of Section 6.86.6. Exception to 6.5: Attached dwelling units and spaces that meet the local ventilation requirements set for bathrooms in Section 5 [of ASHRAE 62.2]. 6.65.1 Habitable Spaces. Each habitable space shall be provided with ventilation openings with an openable area not less than 4% of the floor area or less than 5 ft² (0.5 m²). 6.65.2 Toilets and Utility Rooms. Toilets and utility rooms shall be provided with <u>natural</u> ventilation openings with an openable area not less than 4% of the room floor area or less than 1.5 ft² (0.15 m²). Exceptions to 6.5.2: 1. Utility rooms with a dryer exhaust duct. 2. Toilet compartments in bathrooms.
0706	6.86.6 Air Inlets. Air inlets that are part of the ventilation design shall be located a minimum of 10 ft (3 m) from known sources of contamination such as a stack, vent, exhaust hood, or vehicle exhaust. The intake shall be placed so that entering air is not obstructed by snow, plantings, or other material. Forced air inlets shall be provided with rodent/insect screens (mesh not larger than 1/20.5 in. [13 mm]). Exceptions to 6.6: 1. Ventilation openings in the wall may be as close as a stretched string distance of 3 ft (1 m) from sources of contamination exiting through the roof or dryer exhausts. 2. No minimum separation distance shall be required between windows and local exhaust outlets in kitchens and bathrooms. 3. Vent terminations covered by and meeting the requirements of the National Fuel Gas Code (NFPA 54/ANSI Z223.1)7 or equivalent. 4. Where a combined exhaust/intake termination is used to separate intake air from exhaust air originating +* 0, in a living space other than kitchens, no minimum separation distance between these two openings is required. For these combined terminations, the exhaust air concentration within the intake airflow shall not exceed 10%, as established by the manufacturer.
0807	6.96.8 Carbon Monoxide Alarms. A carbon monoxide alarm shall be installed in each dwelling unit in accordance with NFPA 720, <i>Standard for the Installation of Carbon Monoxide (CO) Detection and Warning Equipment</i> <i>National Fire Alarm and Signaling Code</i>, and shall be consistent with requirements of applicable laws, codes, and standards.
0908	Air Filter Efficiency. Supply only ventilation systems, makeup air systems, and supply side balanced systems including HRV/ERV shall be provided with air filters having a designated efficiency equal to or greater than MERV 13 when tested in accordance with ASHRAE Standard 52.2, or a particle size efficiency rating equal to or greater than 50% in the 0.30-1.0 um range and equal to or greater than 85% in the 1.0-3.0 um range when tested in accordance with AHRI Standard 680.
The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.	

GK. Air Moving Equipment

The items listed below (7.1 through ~~7.47.5~~) correspond to the information given in ASHRAE 62.2 Section 7 "Air-Moving Equipment". Refer also to Chapter 4.6 of the Residential Compliance Manual (Section 4.6.9) for information describing these requirements in more detail. The signature of the Responsible Person in the declaration statement below certifies that the dwelling complies with these requirements specified in ASHRAE 62.2 Section 7.1 through ~~7.47.5~~ if applicable.

<<Static Text - see the top>>

01	7.1 Selection and Installation Ratings. Ventilation devices and equipment serving individual dwelling units shall be tested in accordance with ANSI/ASHRAE Standard 51/AMCA 210, and ANSI/AMCA Standard 300, and rated in accordance with the airflow and sound rating procedures of the Home Ventilating Institute (HVI 915, 916, 920). Installations of systems or equipment shall be carried out in accordance with manufacturers' design requirements and installation instructions. Airflow and sound ratings shall be provided for ventilation devices and equipment serving individual dwelling units. Airflow and sound ratings shall be provided in accordance with HVI 920, or equivalent, by an administration and certification body that is accredited in accordance with ISO/IEC 17065 with respect to application of the standards and test procedures referenced in Section 7.1 and accredited by an accreditation body operating in accordance with ISO/IEC 17011. Laboratory tests of representative units shall be conducted for airflow in accordance with ANSI/ASHRAE Standard 51/AMCA 210, as prescribed by HVI 916, or equivalent, and conducted for sound in accordance with ANSI/AMCA Standard 300, as prescribed by HVI 915, or equivalent. This section does not require certification to HVI 917.
02	7.2 Installation. Installations of systems or equipment shall be carried out in accordance with manufacturer's design requirements and installation instructions.
0203	7.2.2.2 Sound Ratings for Fans. Ventilation fans shall be rated for sound at no less than the minimum airflow rate required by this standard as noted below. These sound ratings shall be at a minimum of 0.1 in. of water (25 Pa) static pressure in accordance with the HVI procedures referenced in Section 7.1. Exception to 7.2: HVAC air handlers and remote mounted fans need not meet sound requirements. To be considered for this exception, a remote mounted fan must be mounted outside the habitable spaces, bathrooms, toilets, and hallways, and there must be at least 4 ft (1 m) of ductwork between the fan and the intake grille. 7.2.3.1 Dwelling Unit Ventilation or Continuous Local Exhaust Fans. These fans shall be rated for sound at a maximum of 1.0 sone. 7.2.3.2 Demand-Controlled Local Exhaust Fans. Bathroom exhaust fans used to comply with Section 5.2 shall be rated for sound at a maximum of 3 sones. Kitchen exhaust fans used to comply with Section 5.2 shall be rated for sound at a maximum of 3 sones at one or more airflow settings greater than or equal to 100 cfm (47 L/s). Exception to 7.2.2: 1. Fans with a minimum airflow setting exceeding 400 cfm (189 L/s) need not comply. 2. Kitchen Range Hoods may be rated for sound at the static pressure determined at working speed as specified in HVI 916 section 7.
0304	7.37.4 Exhaust Ducts. 7.34.1 Multiple Exhaust Fans Using One Duct. Exhaust fans in separate dwelling units shall not share a common exhaust duct. If more than one of the exhaust fans in a single dwelling unit shares a common exhaust duct, each fan shall be equipped with a backdraft damper to prevent the recirculation of exhaust air from one room to another through the exhaust ducting system. 7.34.2 Single Exhaust Fan Ducted to Multiple Exhaust Inlets. Where exhaust inlets are commonly ducted across multiple dwelling units, one or more exhaust fans located downstream of the exhaust inlets shall be designed and intended to run continuously, or a system of one or more backdraft dampers shall be installed to isolate each dwelling unit from the common duct when the fan is not running.
0405	7.47.5 Supply Ducts. Where supply outlets are commonly ducted across multiple dwelling units, one or more supply fans located upstream of all the supply outlets shall be designed and intended to run continuously, or a system of one or more backdraft dampers shall be installed to isolate each dwelling unit from the common duct when the fan is not running.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.